P01

Lumped Thermal Resistance Networks in Electronic Packaging

Authors G. N. Ellison

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Domain DOM_ENG

Abstract

Heat flow through integrated circuits is modeled as a lumped resistor-capacitor network. Temperature differences drive heat current through thermal resistances in direct compliance with Ohm's and Kirchhoff's laws.

Abstract quoted from G. N. Ellison et al., Thermal Computations for Electronic Equipment (1984).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?